

Nabh Garg

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Software Development Engineer driving end-to-end product ownership to solve user problems and build scalable architectures. Metric-driven problem solver skilled in Python, JavaScript, and multi-DB environments, passionate about writing clean code and elevating engineering standards.

Education

VIT Bhopal University

June 2023 – May 2027

B.Tech, Computer Science Engineering – CGPA: 8.2, No Active Backlogs

DAV Kalisindh Thermal Public School

2021 – 2023

Senior Secondary (Class XII) – 75%+

Technical Skills

Backend & Languages: Python, JavaScript, Node.js, Express.js, FastAPI, Flask, Java, C++

Databases: MySQL (Relational Schema Design), MongoDB (NoSQL Aggregations)

Architecture & Performance: End-to-End Ownership, REST API Design, Microservices, Scalability Optimization

Security & Real-Time: WebSockets, WebRTC, Access Control, Modbus, Fault-Tolerant System Design

Core CS: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

Experience

Dholpur Combined Cycle Power Plant (RVUNL)

May 2026 – Jul 2026

Semester Internship – (45 Days, On-Site)

- Studied the GE Mark VIe Distributed Control System (DCS) architecture governing a mission-critical industrial system, focusing on protective trip design and real-time parameter control.
- Analyzed SCADA principles and plant communication protocols (Modbus, EtherNet/IP) underlying fault-tolerant, distributed real-time systems.
- Documented findings connecting large-scale industrial control systems to core backend CS concepts, including real-time processing and industrial networking.

Projects

ELLE – Environmental Lens for Liquid Evaluation

Live | [GitHub](#)

- Engineered end-to-end backend logic for a microplastic detection platform, driving a 90.9% automated accuracy rate to replace manual lab processes.
- Powered by Python, Flask, MongoDB, and MQTT-based IoT telemetry ingestion for real-time sensor updates.
- Drove performance and scalability metrics by resolving a 512MB RAM constraint via an ONNX-headless lazy-loading pipeline, simultaneously cutting CPU usage by 70%.

SehPaathi – AI Exam & Document Assistant

Live | [GitHub](#)

- Owned the design, implementation, and verification of an interactive analysis backend that polls document states for complex queries.
- Architected with Python, FastAPI/Flask logic, Gemini APIs, and secure environment-secret API configurations.
- Elevated system maintainability by architecting isolated session-state management across independent modules, ensuring zero data leakage during concurrent usage.

Certifications

- The Bits and Bytes of Computer Networking** – Coursera (Google), Nov 2024 – [Verify Credential](#)
- Graphs Camp** Completion Certificate (Algo University) – [Verify Credential](#)

Achievements

- 1st Place** – CodeVerse Hackathon 2025 (200+ teams), VIT Bhopal – built a full-stack IoT health platform
- 3rd Place** – Blockchain Nexus, Techkriti'25, IIT Kanpur
- Galactic Problem Solver** – NASA International Space Apps Challenge

Co-Curricular Activities

Mharo Rajasthan Club, VIT Bhopal University

May 2025 – Present

Technical Team Lead

- Mentored and led a technical module to manage operations and cross-team coordination for a flagship university event with 1,500+ entries.

Unnat Bharat Abhiyan (Ministry of Education), VIT Bhopal

Nov 2025 – Present

Innovation & Technology Coordinator

- Develop technology-driven solutions addressing real-world rural community challenges, aligning with initiatives for grassroots livelihood improvement.

Strategic Interests

Chess & System Design

- Avid chess player leveraging strategic foresight, rapid decision-making, and analytical thinking in problem-solving and software architecture.